MIL-S-19500/258A(NAVY)
22 June 1964
SUPERSEDING
MIL-S-19500/258(NAVY)
9 January 1963
(See 6.2)

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR TYPES

2N962 AND 2N964

1. SCOPE

1.1 Description.- This specification covers the detail requirements for PNP germanium epitaxial switching transistors and is in accordance with MIL-S-19500, except as otherwise specified herein.

1.2 Mechanical dimensions and outline.- T018 (see figure 1).

1.3 Absolute maximum ratings.-

P_T (mw)		V_{CB} (Vdc)	
T_C = 25°C. 1/	T_A = 25°C. 2/	2N962	2N964
300	150	-12	-15

V_{EB} (Vdc)		V_{CES} (Vdc)		I_C mA	T_J	T_{stg}
2N962	2N964	2N962	2N964			
-1.25	-2.5	-12	-15	200	100°C.	-55° to 100°C.

1/Derate 4 mw/°C above 25°C.
2/Derate 2 mw/°C above 25°C.

1.4 Primary electrical characteristics.

	h_{FE}				h_{fe}	V_{CE} (sat) (Vdc)			
	V_{CE} = -0.3 I_C = -10 ma		V_{CE} = -1Vdc I_C = -100 ma		f = 100 mc I_C = -20 ma, V V_{CE} = -1Vdc	I_c = -10 madc I_b = -1 madc		I_c = -100 madc I_b = -10 madc	
	2N962	2N964	2N962	2N964		2N962	2N964	2N962	2N964
Max.	20 ---	40 160	20 ---	40 ---	3.0 ---	--- -0.20	--- -0.18	--- -0.70	--- -0.60

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:56Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY
MIL-S-19491 - Semiconductor Devices, Preparation for Delivery of.
MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Transistors shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Design, construction and physical dimensions.- Transistors shall be of the design, construction and physical dimensions shown on figure 1.

3.3 Performance characteristics.- Performance characteristics shall be as specified in 4.2.1 and 4.2.2.

3.4 Marking.- The following marking specified in MIL-S-19500 may be omitted from the device:

(a) Country of origin.
(b) Manufacturer's identification.

3.4.1 The device shall be marked with the "USN" prefix in lieu of the "JAN" prefix.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests.- Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.2.1, and 4.2.2. (Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.1)).

4.2 Quality conformance inspection.- Quality conformance inspection shall consist of the examinations and tests specified in 4.2.1 and 4.2.2.

4.2.1 Group A inspection.- Group A inspection shall consist of the examinations and tests shown in table I.

4.2.2 Group B inspection.- Group B inspection shall consist of the examinations and tests shown in table II.

4.2.2.1 Destructive tests.- Tests listed in subgroups 2, 3, 4, and 5 are considered destructive.

4.2.2.2 Salt atmosphere.- The device shall be examined for destructive corrosion and illegible marking.

4.2.4 Quality conformance inspection information.- When specified in the contract or order, one copy of the quality conformance inspection information pertinent to the transistor inspection lot shall be furnished by the transistor supplier and shall accompany each transistor shipment from the inspection lot to the equipment manufacturer.

4.3 Inspection of preparation for delivery.- Sample items and packs shall be selected and inspected in accordance with MIL-S-19491 to verify conformance with requirements in section 5 of this specification.

4.4 Collector latchup voltage.- Latch-up is the failure of the collector potential to return to the supply voltage upon turn-off. The collector latchup voltage is the minimum voltage that the collector must return to upon turn-off of the transistor.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:56Z
Check the source to verify that this is the current version before use.

4.5 Base leakage current.- I_{BL} is defined as base leakage current with both junctions reverse biased. I_C is always less than I_{BL} for $V_{BE} > V_T$. (V_{BE} is off condition base bias, V_T is base voltage at threshold of conduction.)

4.6 Total control charge.- When a transistor is held in a conductive state by a current, I_{B1}, a charge Q_S is developed in the active region. A charge Q_T can be stored on an external capacitor C to neutralize the internal charge of the transistor.

4.7 Methods of examination and test.-

4.7.1 Input capacitance.- Input capacitance shall be conducted in accordance with Method 3240 of MIL-STD-750 except that the capacitor in the output circuit shall be omitted.

4.7.2 All measurements and tests shall be made at an ambient temperature of 25° ± 3°C, unless otherwise specified.

4.7.3 Shock.- The shock waveform shall approximate one half-cycle of a sinewave with a rise time of $0.25\ ^{+0.1}_{-0}$ msec and a decay time of $0.25\ ^{+0.1}_{-0}$ msec. The acceleration of any secondary impacts shall not exceed 20 percent of each primary impact. The amplitude of any response waveform distortion shall not exceed 10 percent of the peak acceleration.

5. PREPARATION FOR DELIVERY

5.1 See MIL-S-19500.

6. NOTES

6.1 The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.)

6.2 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - Ships
(Project 5960-N461(NAVY))

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:56Z
Check the source to verify that this is the current version before use.

Table I - Group A inspection.

EXAMINATION OR TEST	CONDITIONS		LTPD	MIN. REJ. NO.	SYMBOL	LIMITS		UNIT
	MIL-STD-750 METHOD	SPECIFIC CONDITIONS				MIN.	MAX.	
Subgroup 1								
Visual and mechanical examination	2071		10	5				
Subgroup 2								
Collector-base breakdown voltage	3001 Condition D	I_C = -100uAdc I_E = 0	5	5	BV_{CBO}			
2N962						-12	---	Vdc
2N964						-15	---	Vdc
Collector latch-up voltage (see 4.4)		V_{CC} = -11.5Vdc (see figure 7)			V_{CEXL}	-11.5	---	Vdc
Emitter to base breakdown voltage	3026 Condition D	I_E = -100uAdc I_C = 0			BV_{EBO}			
2N962						-1.25	---	Vdc
2N964						-2.5	---	Vdc
Collector to emitter breakdown voltage	3011 Condition D	I_C = -10mAdc I_B = 0			BV_{CEO}	-7.0	---	Vdc
Collector to emitter cutoff current	3041 Condition C				I_{CES}			
2N962		V_{CE} = -12Vdc V_{EB} = 0				---	-100	µAdc
2N964		V_{CE} = -15Vdc V_{EB} = 0				---	-100	µAdc
Collector to base cutoff current	3036 Condition D	V_{CB} = -6Vdc I_E = 0			I_{CBO}	---	-3.0	µAdc
Base leakage current (see 4.5)		(See figure 6) V_{CE} = -6Vdc V_{BE} = +0.5Vdc			I_{BL}	---	4	µAdc
		V_{CE} = -6Vdc V_{BE} = +0.5Vdc T_J = +85°C.			I_{BL}	---	140	µAdc
Subgroup 3								
Static forward current transfer ratio	3076	I_C = -10mAdc V_{CE} = -0.3Vdc			h_{FE}			
2N962						20	---	
2N964						40	160	

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:56Z
Check the source to verify that this is the current version before use.

Table I - Group A inspection (Cont'd).

EXAMINATION OR TEST	CONDITIONS		LTPD	MIN. REJ. NO.	SYMBOL	LIMITS		UNIT
	MIL-STD-750 METHOD	SPECIFIC CONDITIONS				MIN.	MAX.	
Static forward-current transfer ratio	3076	I_C = -50mAdc V_{CE} = -1Vdc			h_{FE}			
2N962						20	---	---
2N964						40	---	---
Static forward-current transfer ratio	3076	I_C = -100mAdc V_{CE} = -1Vdc			h_{FE}			
2N962						20	---	---
2N964						40	---	---
Collector saturation voltage	3071	I_C = -10mAdc I_B = -1mAdc			$V_{CE(sat)}$			
2N962						---	-0.20	Vdc
2N964						---	-0.18	Vdc
Collector saturation voltage	3071	I_C = -50mAdc I_B = -5mAdc			$V_{CE(sat)}$			
2N962						---	-0.40	Vdc
2N964						---	-0.35	Vdc
Collector saturation voltage	3071	I_C = -100mAdc I_B = -10mAdc			$V_{CE(sat)}$	---		

2N962						---	-0.70	Vdc
2N964						---	-0.60	Vdc
Base saturation voltage	3066 Condition A	I_C = -10mAdc I_B = -1mAdc			$V_{BE(sat)}$	-0.30	-0.50	Vdc
Base saturation voltage	3066 Condition A	I_C = -50mAdc I_B = -5mAdc			$V_{BE(sat)}$	-0.40	-0.75	Vdc
Base saturation voltage	3066 Condition A	I_C = -100mAdc I_B = -10mAdc			$V_{BE(sat)}$			
2N962						-0.40	-1.25	Vdc
2N964						-0.40	-1.00	Vdc
Subgroup 4								
Small-signal short circuit forward-current transfer ratio	3206	I_E = -20mAdc V_{CE} = -1.0 Vdc f = 100 mc	5	5	h_{fe}	3	---	---

Source: https://assist.dla.mil -- Downloaded: 2026l0-18T13:56Z
Check the source to verify that this is the currentversion before use.

Table I - Group A inspection (Cont'd).

EXAMINATION OR TEST	CONDITIONS		LTPD	MIN. REJ. NO.	SYMBOL	LIMITS		UNIT
	MIL-STD-750 METHOD	SPECIFIC CONDITIONS				MIN.	MAX.	
Input capacitance (output open-circuited	3240 (See 4.7)	V_{EB} = -1Vdc I_C = 0 f = 100 kc			C_{ibo}	---	3.5	pf
Collector capacitance	3236	V_{CB} = -1Vdc I_E = 0 f = 1 mc			C_{obo}	---	5.0	pf
Subgroup 5								
Delay plus rise time		I_C = -10mAdc I_{B1} = 1mAdc $V_{BE(0)}$ = +1.25Vdc (see figure 2)	5	5	$t_d + t_r$	---	50	nsec
Delay plus rise time		I_C = -100mAdc I_{B1} = -5mAdc $V_{BE(0)}$ = 1.25Vdc (see figure 3)			$t_d + t_r$	---	50	nsec
Storage plus fall time		I_C = -10mAdc I_{B1} = -1mAdc I_{B2} = 0.25mAdc $V_{BE(0)}$ = 1.25Vdc (see figure 2)			$t_s + t_f$			
2N962						---	100	nsec
2N964						---	85	nsec
Storage plus fall time		I_C = -100mAdc I_{B1} = -5mAdc I_{B2} = 1.25mAdc $V_{BE(0)}$ = 1.25mAdc (see figure 3)			$t_s + t_f$			
2N962						---	180	nsec
2N964						---	85	nsec
Total control charge		I_C = -10mAdc I_B = -1mAdc (see figure 4)			QT			
2N962						---	90	pc
2N964						---	80	pc
Total control charge (see 4.6)		I_C = -100mAdc I_B = -5mAdc (see figure 5)			QT			
2N962						---	150	pc
2N964						---	125	pc

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:56Z
Check the source to verify that this is the current version before use.

Table I - Group A inspection (Contd).

EXAMINATION OR TEST	CONDITIONS		LTPD	MIN. REJ. NO.,	SYMBOL	LIMITS		UNIT
	MIL-STD-750 METHOD	SPECIFIC CONDITIONS				MIN.	MAX.	
Subgroup 6								
Collector to base cutoff current	3036 Condition D	T_A = +55°C V_{CB} = -6Vdc I_E = 0	20	4	I_{CBO}	---	-24	µAdc
Static forward-current transfer ratio	3076	T_A = -55°C I_C = -10mAdc V_{CE} = -0.3Vdc			h_{FE}			
2N962						10	---	---
2N964						20	---	---
Static forward-current transfer ratio	3076	T_A = +85°C I_C = -100mAdc V_{CE} = -1Vdc			h_{FE}			
2N962						18	---	---
2N964						35	---	---

Table II - Group B inspection.

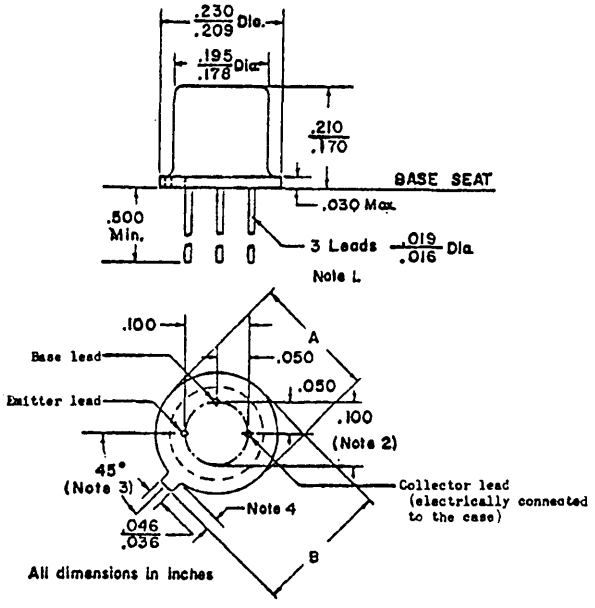
EXAMINATION OR TEST	CONDITIONS		LTPD	MIN. REJ. NO.	SYMBOL	LIMITS		UNIT
	MIL-STD-750 METHOD	SPECIFIC CONDITIONS				MIN.	MAX.	
Subgroup 1								
Physical dimensions	2066		20	4				
Subgroup 2								
Solderability	2026		10	5				
Thermal shock (temperature cycling)	1051 Condition A							
Thermal shock (glass strain)	1056 Condition A							
Moisture resistance	1021							
Subgroup 3								
Shock (see 4.7.3)	2016	Nonoperating 1500G min. 5 blows each orientation: Y_1, Y_2, X_1 and Z_1	10	5				
Vibration fatigue	2046	Nonoperating						
Vibration, variable frequency	2056							

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:56Z
Check the source to verify that this is the current version before use.

Table II - Group B inspection (Contd).

EXAMINATION OR TEST	CONDITIONS		LTPD	MIN. REJ. NO.	SYMBOL	LIMITS		UNIT
	MIL-STD-750 METHOD	SPECIFIC CONDITIONS				MIN.	MAX.	
Constant acceleration	2006	20,000G each orientation: $\bar{Y}_1$, Y_2, X_1 and Z_1						
Subgroup 4								
Terminal strength	2036 Condition E		20	4				
Subgroup 5			20	4				
Salt atmosphere (corrosion)	1041							
Subgroup 6								
High temperature life (nonoperating)	1031	T_A = 100°C min.	λ = 10	5				
Subgroup 7								
Steady state operation life	1026	T_A = 25°C V_{CB} = -5Vdc I_C = -30mAdc	λ = 5	5				
Endpoints for subgroups 2, 3, 5, 6, and 7:								
Static forward-current transfer ratio	3076	I_C = 10mAdc V_{CE} = -0.3Vdc	---	---	h_{FE}			
2N962						16	---	---
2N964						32	---	---
Collector to base cutoff current	3036 Condition D	V_{CE} = -6Vdc I_E = 0	---	---	I_{CBO}	---	-6.0	µAdc

Source: https://assist.dla.mil -- Downloaded: 202610-18T13:56Z
Check the source to verify that this is the current version before use.

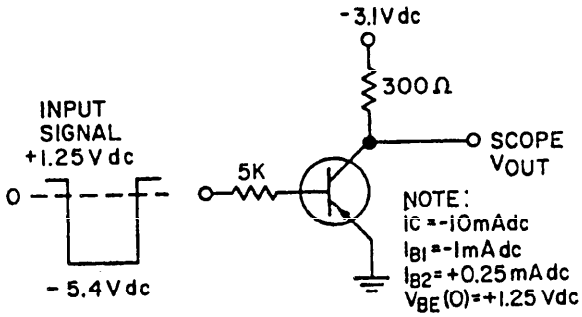


Notes:

1. The specified lead diameter applies to the zone between 0.050 and 0.250 from the base seat. Between 0.250 and end of lead a maximum of 0.021 is held. Outside of these zones the lead diameter is not controlled.
2. Maximum diameter leads at a gaging plane $0.054^{+0.054}_{-0.000}$ below base seat to be within 0.007 of their true location relative to maximum width tab and to the maximum 0.230 diameter measured with a suitable gage. When gage is used, measurement will be made at base seat.
3. Index tab for visual orientation only.
4. Tab length shall be 0.028 minimum, 0.048 maximum, and will be determined by subtracting diameter A from Dimension B.

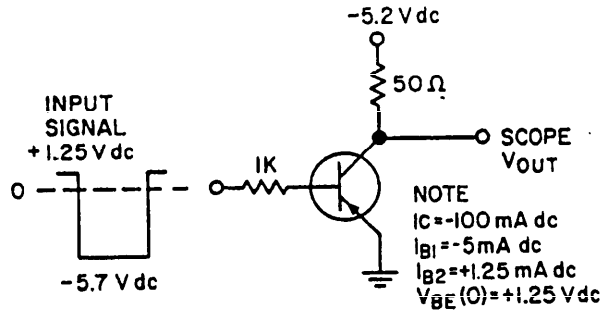
FIGURE 1 - MECHANICAL DIMENSIONS AND OUTLINE FOR TRANSISTOR TYPES 2N962 AND 2N964.

Source: https://assist.dla.mil -- Downloaded: 20261O-18T13:56Z
Check the source to verify that this is the current version before use.



SCOPE INPUT IMPEDANCE ≥ 1MΩ
SCOPE INPUT CAPACITANCE ≤ 20pf
GENERATOR OUTPUT IMPEDANCE = 50Ω
INPUT PULSE tr = tf ≤ 2nsec
SCOPE RISE TIME ≤ 10sec
INPUT PULSE WIDTH = 8 msec (50% DUTY CYCLE)

FIGURE 2 - 10mA (Ic) SWITCHING TIME TEST CIRCUIT



SCOPE INPUT IMPEDANCE ≥ 1MΩ
SCOPE INPUT CAPACITANCE ≤ 20pf
GENERATOR OUTPUT IMPEDANCE = 50Ω
INPUT PULSE tr = tf ≤ 2nsec
SCOPE RISE TIME ≤ 10nsec
INPUT PULSE WIDTH = 8 msec (50% DUTY CYCLE)

FIGURE 3 - 100mA(Ic) SWITCHING TIME TEST CIRCUIT

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:56Z
Check the source to verify that this is the current version before use.

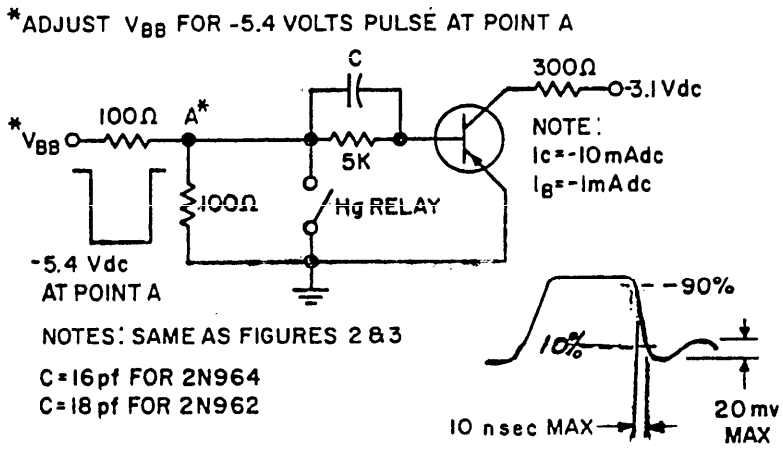


FIGURE 4 - 10mA (Ic) TOTAL CONTROL CHARGE TEST CIRCUIT

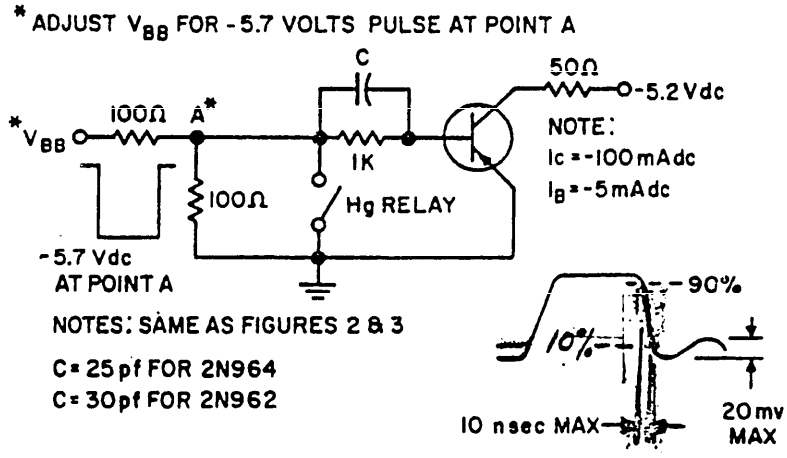


FIGURE 5 - 100mA (Ic) TOTAL CONTROL CHARGE TEST CIRCUIT

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T13:56Z
Check the source to verify that this is the current version before use.

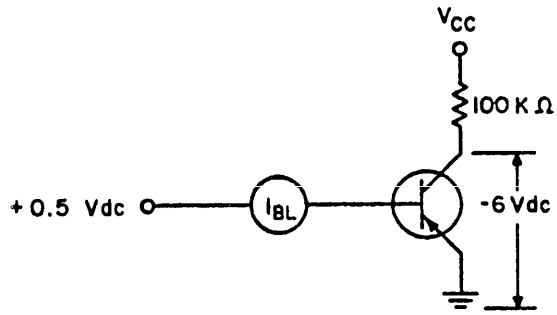
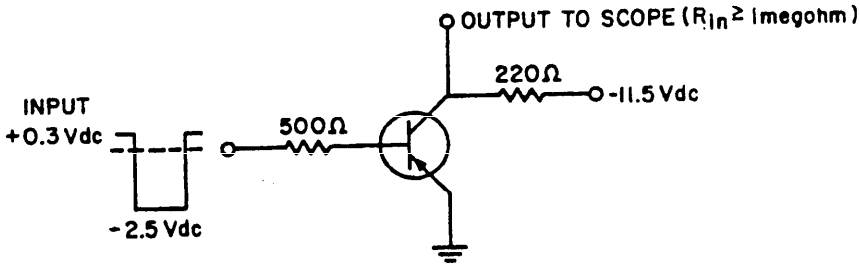


FIGURE 6 – BASE LEAKAGE CURRENT TEST CIRCUIT



NOTE:
GENERATOR OUTPUT IMPEDANCE ≤ 50 Ω
INPUT PULSE: $t_r = t_f \leq 2$nsec, tp ≥ 500 nsec, DUTY CYCLE ≤ 50%

FIGURE 7 – COLLECTOR LATCH-UP VOLTAGE TEST CIRCUIT

Source: https://assist.dla.mil -- Downloaded: 20261O-18T13:56Z
Check the source to verify that this is the current version before use.